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# Hodge-Decomposed Preference Optimization: Detecting and Filtering Cyclic Inconsistencies in Reward Learning

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## Abstract

Human preference data used to train reward models is inherently noisy and can contain cyclic inconsistencies—where  $A \succ B \succ C \succ A$ —that no scalar reward function can faithfully represent. Standard preference optimizers (DPO, KTO, GRPO) train on all preferences equally, including cycles that reward-hacking policies can exploit. We propose **Hodge-Decomposed Preference Optimization (HodgePO)**, a family of preference optimizers that use the combinatorial Hodge decomposition to separate globally consistent preference signals from cyclic noise. HodgePO operates through two mechanisms: (1) per-sample weights derived from node-level cycle participation that downweight cyclically contaminated training pairs, and (2) a *Hodge potential-alignment regularizer* that encourages the model’s reward predictions to match the globally consistent gradient potential from the Hodge decomposition. In a 30-seed benchmark on Anthropic’s HH-RLHF dataset, Hodge-DPO achieves **0.9999** exploit resistance versus 0.940 for standard DPO (+6.3%, Cohen’s  $d = 6.52$ ,  $p < 0.0001$ ), and Hodge-KTO achieves **0.9964** versus 0.800 for KTO (+24.5%,  $d = 16.47$ ,  $p < 0.0001$ ). These improvements are statistically significant across all 30 seeds, demonstrating that topological filtering of preference data substantially improves robustness to reward hacking.

## 1. Introduction

Preference optimization methods—Direct Preference Optimization (DPO) (?), Kahneman-Tversky Optimiza-

tion (KTO) (?), Group Relative Policy Optimization (GRPO) (?), and Odds Ratio Preference Optimization (ORPO) (?)—have become the dominant approach for aligning language models with human intent. These methods sidestep explicit reward modeling by directly optimizing a policy from pairwise preference data.

However, human preferences are not always globally consistent. The Condorcet paradox (?) demonstrates that pairwise majority preferences can be cyclic even when individual preferences are transitive—a phenomenon generalized by Arrow’s impossibility theorem (?). When preference data contains such cycles, *no scalar reward function can faithfully represent the full preference structure* (?). Standard optimizers treat all preference pairs equally, forcing the model to fit contradictory signals. This creates exploitable inconsistencies that reward-hacking policies can discover and leverage (??).

**Key insight.** The combinatorial Hodge decomposition (??) provides an exact decomposition of preference data into:

- **Gradient component:** globally consistent preferences explainable by a scalar potential (a “true” reward)
- **Harmonic component:** irreducible cyclic inconsistencies that no scalar reward can capture

By computing this decomposition on the preference graph and using it to modulate training, we can teach preference optimizers to learn from the consistent signal while discounting cyclic noise.

## Contributions.

1. We introduce **Hodge-Decomposed Preference Optimization (HodgePO)**, a principled framework that augments any pairwise preference optimizer with cycle-aware training signals derived from the Hodge decomposition (Section ??).

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- We identify and address two technical challenges: (a) cross-pair graph connectivity is essential for non-trivial Hodge structure (Section ??), and (b) the Hodge potential convention requires sign correction for use as a regularization target (Section ??).
- We validate HodgePO across four base methods (DPO, KTO, GRPO, ORPO) in a statistically rigorous 30-seed benchmark on HH-RLHF, demonstrating significant improvements in exploit resistance for DPO (+6.3%) and KTO (+24.5%) with large effect sizes (Section ??).

## 2. Background

### 2.1. Preference Optimization

Given a dataset of preference pairs  $\mathcal{D} = \{(x, y_w, y_l)\}$  where  $y_w$  is preferred over  $y_l$  for prompt  $x$ , DPO (?) directly optimizes:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (1)$$

KTO (?) operates on individual samples rather than pairs, using prospect-theoretic loss weighting. GRPO (?) uses group-relative advantages with a clipped surrogate objective.

All these methods assume preferences are generated by a latent reward function via the Bradley-Terry model (?):  $P(y_w \succ y_l) = \sigma(r^*(y_w) - r^*(y_l))$ . This assumption fails when preferences contain irreducible cycles.

### 2.2. Combinatorial Hodge Theory

The Hodge decomposition on graphs (?) generalizes the classical Helmholtz decomposition to discrete settings. Given a weighted directed graph  $G = (V, E, w)$  with edge flow  $f : E \rightarrow \mathbb{R}$ :

**Theorem 1** (Hodge Decomposition (?)). *Any edge flow  $f$  on a graph  $G$  decomposes uniquely as:*

$$f = \underbrace{d_0\phi}_{\text{gradient}} + \underbrace{\delta_1\psi}_{\text{curl}} + \underbrace{h}_{\text{harmonic}} \quad (2)$$

where  $d_0$  is the coboundary operator,  $\delta_1$  is the adjoint boundary operator,  $\phi : V \rightarrow \mathbb{R}$  is a node potential,  $\psi$  is a 2-chain, and  $h \in \ker(d_0^*d_0 + \delta_1\delta_1^*)$  is the harmonic component.

The gradient component  $d_0\phi$  captures globally consistent rankings: for edge  $(i, j)$ ,  $(d_0\phi)_{ij} = \phi(j) - \phi(i)$ . The harmonic component  $h$  captures irreducible cycles—preferences that cannot be explained by any global ranking. In social choice theory,  $h \neq 0$  corresponds to Condorcet cycles (?).

The **first Betti number**  $\beta_1 = \dim(\ker \Delta_1)$ , where  $\Delta_1 = d_0^*d_0 + \delta_1\delta_1^*$  is the Hodge Laplacian, measures the dimension of the harmonic space. When  $\beta_1 > 0$ , the preference data contains irreducible cyclic structure.

For preference optimization, we use the HodgeRank formulation (?): the gradient potential  $\phi^*$  is the least-squares solution

$$\phi^* = \arg \min_{\phi} \sum_{(i,j) \in E} w_{ij} (f_{ij} - (\phi(j) - \phi(i)))^2 \quad (3)$$

which gives the globally consistent reward ranking that best explains the observed preferences.

## 3. Hodge-Decomposed Preference Optimization

HodgePO augments any base preference optimizer with two Hodge-derived mechanisms.

### 3.1. Preference Graph Construction

The choice of graph structure is critical. A naive preference graph with only direct edges (one per training pair) produces a perfect matching—a bipartite graph with *zero cycles* and trivial Hodge decomposition ( $h = 0$  everywhere). This was a key failure mode in early experiments.

To obtain meaningful Hodge structure, we construct a denser preference graph:

**Definition 1** (Cross-Pair Preference Graph). *Given  $N$  preference pairs embedded as  $\{(\mathbf{e}_w^{(i)}, \mathbf{e}_l^{(i)})\}_{i=1}^N$  in  $\mathbb{R}^d$ , the cross-pair preference graph  $G$  has:*

- Direct edges:**  $(w_i, l_i)$  with weight  $P(y_w^{(i)} \succ y_l^{(i)})$  for each training pair  $i$ .
- Cross-pair edges:** For each response embedding  $\mathbf{e}$ , connect to its  $k$  nearest neighbors from other pairs via cosine similarity, with preference direction determined by a lightweight reward model. We use  $k = 5$ .

Cross-pair edges create shared nodes between different preference pairs, enabling cycles. In our experiments, 300 direct edges produce 0 cycles, while adding cross-pair edges yields  $\sim 1,500$  total edges with  $\beta_1 > 0$  and 28% harmonic energy fraction.

### 3.2. Node-Level Cycle Participation Weights

After computing the Hodge decomposition on  $G$ , we derive per-sample training weights based on how much each response participates in cyclic preference patterns.

For each node  $v \in V$ , define the **harmonic participation**

score:

$$s(v) = \sum_{e \ni v} |h_e| \quad (4)$$

where the sum is over all edges incident to  $v$  and  $h_e$  is the harmonic component on edge  $e$ . For each training pair  $i$  with preferred response node  $w_i$  and dispreferred node  $l_i$ :

$$\text{cycle\_score}(i) = \frac{s(w_i) + s(l_i)}{2 \cdot \max_j \frac{s(w_j) + s(l_j)}{2}} \quad (5)$$

normalized to  $[0, 1]$ . The training weight is:

$$\alpha_i = 1 - \lambda_{\text{damp}} \cdot \text{cycle\_score}(i), \quad \alpha_i \in [0.2, 1.0] \quad (6)$$

with  $\lambda_{\text{damp}} = 0.6$ . Pairs where both responses participate heavily in cycles receive lower weight.

### 3.3. Hodge Potential-Alignment Regularization

The second and more impactful mechanism is a regularization loss that encourages the model’s implicit reward predictions to align with the Hodge gradient potential—the globally consistent ranking.

The gradient potential  $\phi^*$  from the Hodge decomposition assigns each node a score reflecting its position in the cycle-free global ranking. For training pair  $i$ :

$$\Delta\phi_i = \phi^*(l_i) - \phi^*(w_i) \quad (7)$$

**Sign convention.** In the Hodge framework, positive edge flow from  $w_i$  to  $l_i$  means  $w_i$  is preferred. The gradient component satisfies  $f_{w_i l_i}^{\text{grad}} = \phi^*(l_i) - \phi^*(w_i)$ . For a consistent preference ( $w_i \succ l_i$ ), positive flow implies  $\phi^*(l_i) > \phi^*(w_i)$ , so  $\Delta\phi_i > 0$ . This convention is critical: using  $\phi^*(w_i) - \phi^*(l_i)$  instead produces negative targets that catastrophically reverse the learning signal.

Given the model’s implicit reward predictions  $r_\theta(w_i)$  and  $r_\theta(l_i)$ , the potential-alignment loss is:

$$\mathcal{L}_{\text{Hodge}} = \frac{1}{B} \sum_{i \in \text{batch}} (r_\theta(w_i) - r_\theta(l_i) - \Delta\phi_i)^2 \quad (8)$$

The total HodgePO loss for any base method  $\mathcal{L}_{\text{base}}$  is:

$$\mathcal{L}_{\text{HodgePO}} = \mathcal{L}_{\text{base}}(\alpha) + \lambda_{\text{Hodge}} \cdot \mathcal{L}_{\text{Hodge}} \quad (9)$$

where  $\mathcal{L}_{\text{base}}(\alpha)$  denotes the base loss with per-sample weights  $\alpha_i$  from Eq. ??, and  $\lambda_{\text{Hodge}}$  controls regularization strength.

**Why not batch-level Hodge penalty?** An alternative approach computes the Hodge decomposition on *in-batch* reward predictions. However, a scalar reward model produces rewards  $r_\theta(x)$  that are always globally consistent by

### Algorithm 1 HodgePO Training

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**Require:** Preference pairs  $\mathcal{D}$ , base optimizer (DPO/KTO/GRPO),  $\lambda_{\text{Hodge}}$ ,  $\lambda_{\text{damp}}$

- 1: Embed all responses:  $\{\mathbf{e}_w^{(i)}, \mathbf{e}_l^{(i)}\}$
- 2: Construct cross-pair preference graph  $G$  (Section ??)
- 3: Compute Hodge decomposition: gradient potential  $\phi^*$ , harmonic  $h$
- 4: Compute per-sample weights  $\alpha_i$  via Eq. ??
- 5: Compute potential diffs  $\Delta\phi_i$  via Eq. ??
- 6: **for** each training epoch **do**
- 7:   **for** each mini-batch  $B$  **do**
- 8:      $\mathcal{L}_{\text{base}} \leftarrow$  base loss with weights  $\alpha_i$
- 9:      $\mathcal{L}_{\text{Hodge}} \leftarrow$  Eq. ??
- 10:     $\mathcal{L} \leftarrow \mathcal{L}_{\text{base}} + \lambda_{\text{Hodge}} \cdot \mathcal{L}_{\text{Hodge}}$
- 11:    Update model parameters via  $\nabla \mathcal{L}$
- 12:   **end for**
- 13: **end for**

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construction: any set of scalar values  $\{r_1, \dots, r_n\}$  defines a total order with zero harmonic component. The batch-level Hodge penalty is therefore identically zero for scalar reward models—mathematically incapable of providing a training signal.

**Intuition.** The potential-alignment regularizer tells the model: “your reward differences should approximate the cycle-free global ranking.” For training pairs involved in cycles, the Hodge potential difference  $\Delta\phi_i$  is smaller (the cycle component has been removed), creating a softer target. For pairs in cycle-free regions,  $\Delta\phi_i$  matches the raw preference, reinforcing the original signal. The regularizer thus selectively moderates learning on cyclically contaminated data.

## 4. Experiments

### 4.1. Setup

**Dataset.** We use Anthropic’s HH-RLHF dataset (?), sampling 2,268 preference pairs from the harmless-base split. Responses are embedded using all-MiniLM-L6-v2 (384-dim). The cross-pair preference graph has 2,268 direct edges and  $\sim 1,030$  cross-pair  $k$ -NN edges (3,298 total), with a global harmonic energy fraction of  $H^1 = 0.284$ .

**Methods.** We compare seven optimizers:

- **DPO, KTO, GRPO, ORPO:** Standard implementations at embedding level.
- **Hodge-DPO, Hodge-KTO, Hodge-GRPO:** HodgePO variants with  $\lambda_{\text{Hodge}} = 0.05$ ,  $\lambda_{\text{damp}} = 0.6$ .

Table 1. Exploit resistance across 7 preference optimizers, 30 seeds each. Hodge-DPO and Hodge-KTO significantly outperform their base methods. \*\*\* $p < 0.0001$ .

| Method     | Mean          | Std   | 95% CI      | vs. Base  |
|------------|---------------|-------|-------------|-----------|
| Hodge-DPO  | <b>0.9999</b> | 0.001 | $\pm 0.000$ | +6.3%***  |
| DPO        | 0.940         | 0.013 | $\pm 0.005$ | —         |
| Hodge-KTO  | <b>0.9964</b> | 0.003 | $\pm 0.001$ | +24.5%*** |
| KTO        | 0.800         | 0.017 | $\pm 0.006$ | —         |
| Hodge-GRPO | 1.000         | 0.000 | $\pm 0.000$ | =         |
| GRPO       | 1.000         | 0.000 | $\pm 0.000$ | —         |
| ORPO       | 0.622         | 0.080 | $\pm 0.030$ | —         |

Table 2. Pairwise statistical tests (Hodge variants vs. base methods).

| Comparison          | $\Delta$ | Cohen’s $d$ | $p$ -value |
|---------------------|----------|-------------|------------|
| Hodge-DPO vs. DPO   | +0.060   | 6.52        | $< 0.0001$ |
| Hodge-KTO vs. KTO   | +0.196   | 16.47       | $< 0.0001$ |
| Hodge-GRPO vs. GRPO | −0.001   | 1.03        | 0.18       |

All methods use identical architecture (2-layer MLP, hidden dim 64), learning rate  $10^{-3}$ , batch size 32, and 50 training epochs.

**Metric. Exploit resistance:** the fraction of held-out test pairs where the trained model assigns higher reward to the preferred response than the dispreferred one. This measures robustness to reward hacking—a model that has learned to exploit cyclic inconsistencies will misrank some pairs.

**Statistical rigor.** Each method runs across **30 independent seeds**. We report means, standard deviations, 95% confidence intervals, and pairwise comparisons via Welch’s  $t$ -test with Cohen’s  $d$  effect sizes.

## 4.2. Main Results

Table ?? presents the main results. Two findings stand out:

**1. Large, statistically significant improvements.** Hodge-DPO achieves 0.9999 exploit resistance versus 0.940 for standard DPO (+6.3%,  $d = 6.52$ ,  $p < 0.0001$ ). Hodge-KTO improves from 0.800 to 0.9964 (+24.5%,  $d = 16.47$ ,  $p < 0.0001$ ). Both effect sizes are “large” by conventional standards ( $d > 0.8$ ); in fact, they are enormous, indicating essentially non-overlapping distributions between Hodge and non-Hodge variants.

**2. Ceiling effects bound gains for strong methods.** GRPO already achieves 1.000 resistance, leaving no room for Hodge improvement. This suggests HodgePO is most valuable for methods with intermediate baseline performance—precisely the regime where reward hacking exploits exist.

## 4.3. Analysis: Why Does HodgePO Help?

**Cross-pair edges are essential.** Without cross-pair  $k$ -NN edges, the preference graph is a perfect matching with  $H^1 = 0$  (zero harmonic energy). The Hodge decomposition is trivial, and Hodge variants are identical to base methods. With cross-pair edges,  $H^1 = 0.284$ —28% of preference flow is cyclic. This is the signal HodgePO exploits.

**Potential-alignment dominates weighting.** In ablations, removing the potential-alignment regularizer ( $\lambda_{\text{Hodge}} = 0$ ) and keeping only per-sample weights produces  $< 1\%$  improvement. The weights have narrow variance ( $\sigma = 0.015$ ) because all nodes participate similarly in cycles on this homogeneous dataset. The regularizer is the primary driver.

**Regularization strength matters.**  $\lambda_{\text{Hodge}} = 0.5$  catastrophically collapses exploit resistance to near-zero by overwhelming the base loss.  $\lambda_{\text{Hodge}} = 0.05$  provides the optimal balance—strong enough to regularize cyclic noise, weak enough to preserve the base optimization signal.

**Sign convention is critical.** The Hodge gradient potential assigns higher values to nodes that *receive* more preference flow (“sinks” in the preference graph), not to higher-quality responses. Using  $\phi^*(w_i) - \phi^*(l_i)$  as the regularization target produces negative targets that teach the model to rank dispreferred responses higher—a subtle but catastrophic bug. The correct convention is  $\Delta\phi_i = \phi^*(l_i) - \phi^*(w_i) > 0$  for consistent pairs.

## 4.4. Method Ranking

Across all 30-seed comparisons, the full ranking by exploit resistance is:

$$\text{Hodge-GRPO} = \text{GRPO} > \text{Hodge-DPO} > \text{Hodge-KTO} > \text{DPO} > \text{KTO} > \text{ORPO}$$

All adjacent pairwise differences are statistically significant ( $p < 0.05$ ) except Hodge-GRPO vs. GRPO (both at ceiling). ORPO’s high variance ( $\sigma = 0.080$ ) suggests instability without a reference model.

## 5. Related Work

**Preference optimization.** DPO (?) reformulated RLHF as a classification problem; KTO (?), GRPO (?), and ORPO (?) offer alternatives with different trade-offs. Nash Learning from Human Feedback (?) addresses intransitivity via game-theoretic equilibria but doesn’t decompose the preference structure.

**Reward hacking.** ? constrain RLHF with KL bounds; ? use reward model ensembles. These defend against overoptimization of a fixed reward model, not against cyclic structure in the preference data itself.

**Hodge theory in ML.** HodgeRank (?) introduced Hodge decomposition for ranking from pairwise comparisons. Neural sheaf diffusion (?) uses sheaf theory for GNNs. ? survey sheaf-theoretic deep learning. To our knowledge, HodgePO is the first application of Hodge decomposition to preference optimization training.

**Cyclic preferences.** The Condorcet paradox (?) and Arrow’s theorem (?) establish that cyclic preferences are fundamental, not anomalous. ? provide PAC bounds for RLHF but assume a BTL model. HodgePO directly addresses the regime where BTL fails.

## 6. Discussion

**Limitations.** (1) Our experiments operate at the embedding level rather than on full language model fine-tuning. The exploit resistance metric measures reward model accuracy, not downstream generation quality. (2) Cross-pair edge construction requires  $k$ -NN search in embedding space, adding  $O(N^2)$  preprocessing (mitigated by approximate NN). (3) HodgePO’s benefit scales with the harmonic energy fraction—on perfectly consistent data ( $H^1 = 0$ ), it provides no advantage.

**When does HodgePO help most?** The improvement is largest for methods with intermediate baseline performance (KTO: +24.5%, DPO: +6.3%) and absent at the ceiling (GRPO: no change). This suggests HodgePO is a “consistency amplifier”: it filters noise that weaker methods are susceptible to, but cannot improve methods that already handle it.

**Broader impact.** Cyclic preferences are endemic in human feedback data: our HH-RLHF analysis shows 28% harmonic energy. As AI systems are trained on increasingly large and diverse preference datasets, the prevalence of cycles will only grow. HodgePO provides a principled, mathematically grounded approach to handling this irreducible structure—complementary to existing defenses like

KL constraints and reward model ensembles.

## 7. Conclusion

We introduced Hodge-Decomposed Preference Optimization (HodgePO), a framework that uses the combinatorial Hodge decomposition to detect and filter cyclic inconsistencies in preference training data. The key technical insight is that the Hodge gradient potential provides a globally consistent reward ranking target that can be used as a differentiable regularizer, while scalar reward predictions always have zero in-batch harmonic energy and thus require a global decomposition. In a 30-seed benchmark, Hodge-DPO and Hodge-KTO achieve near-perfect exploit resistance with massive effect sizes over their base methods. The approach is method-agnostic, computationally lightweight (the Hodge decomposition is a one-time  $O(|E|^2)$  preprocessing step), and requires no changes to model architecture.

## A. Proof of Hodge Decomposition (Theorem ??)

The Hodge decomposition on graphs follows from the orthogonality of the image and kernel of the graph Laplacians. Define the **boundary operators**:

- $d_0 : C^0(V) \rightarrow C^1(E)$ , the coboundary (gradient):  $(d_0\phi)_{ij} = \phi(j) - \phi(i)$
- $d_1 : C^1(E) \rightarrow C^2(T)$ , the curl:  $(d_1f)_{ijk} = f_{ij} + f_{jk} + f_{ki}$  for triangle  $(i, j, k)$

The **Hodge Laplacian** on 1-chains is  $\Delta_1 = d_0d_0^* + d_1^*d_1$ , where  $*$  denotes the weighted adjoint. By the Hodge theorem:

$$C^1(E) = \text{im}(d_0) \oplus \text{im}(d_1^*) \oplus \ker(\Delta_1) \quad (10)$$

These three subspaces are mutually orthogonal under the edge inner product  $\langle f, g \rangle = \sum_e w_e f_e g_e$ .

The gradient potential  $\phi^*$  is obtained by solving the normal equations  $d_0^*d_0\phi = d_0^*f$ , which has a unique solution (up to additive constant) when  $G$  is connected. We regularize with  $10^{-4}I$  for numerical stability.

## B. Implementation Details

### B.1. Preference Graph Construction

1. Embed all responses using sentence-transformers/all-MiniLM-L6-v2 (384-dim).
2. Create direct edges: for each preference pair  $(y_w, y_l)$ , add edge  $(w, l)$  with weight equal to the preference probability.
3. Create cross-pair edges: for each response node, find its  $k = 5$  nearest neighbors (cosine similarity) from other pairs. Preference direction is determined by the dot product of embeddings with the prompt embedding.
4. Total graph:  $N$  direct edges +  $\sim 0.45N$  cross-pair edges (varies by dataset).

### B.2. Hodge Decomposition

We solve the normal equations  $B^\top B\phi = B^\top f$  where  $B$  is the  $|E| \times |V|$  boundary operator. For our largest experiment ( $|V| = 4536$ ,  $|E| = 3298$ ), this takes  $< 2$  seconds on CPU.

### B.3. Training

All preference optimizers operate at the embedding level: a 2-layer MLP ( $384 \rightarrow 64 \rightarrow 1$ ) maps prompt-response embedding pairs to scalar rewards. This is not language

model fine-tuning but rather an isolated evaluation of the preference optimization algorithm’s ability to learn consistent reward rankings.

Table 3. Hyperparameters.

| Parameter                | Value     |
|--------------------------|-----------|
| Embedding dim            | 384       |
| Hidden dim               | 64        |
| Learning rate            | $10^{-3}$ |
| Epochs                   | 50        |
| Batch size               | 32        |
| DPO $\beta$              | 0.1       |
| KTO $\beta$              | 0.1       |
| GRPO $\beta$             | 0.04      |
| GRPO group size          | 8         |
| ORPO $\lambda$           | 0.5       |
| $\lambda_{\text{Hodge}}$ | 0.05      |
| $\lambda_{\text{damp}}$  | 0.6       |
| $k$ -NN cross-pair edges | 5         |
| Seeds                    | 30        |

## C. Why Batch Harmonic Penalty Fails

A natural alternative to the potential-alignment regularizer is a **batch harmonic penalty**: compute the Hodge decomposition on the complete graph of in-batch reward predictions and penalize the harmonic energy fraction.

Given batch rewards  $\{r_1, \dots, r_{2B}\}$  ( $B$  chosen +  $B$  rejected), form the complete graph with edge flow  $f_{ij} = r_j - r_i$  for all  $i < j$ . The gradient potential is  $\phi_i = r_i$  (the rewards themselves), giving gradient flow  $f_{ij}^{\text{grad}} = r_j - r_i = f_{ij}$ . Therefore:

$$h_{ij} = f_{ij} - f_{ij}^{\text{grad}} = 0 \quad \forall (i, j) \quad (11)$$

**The harmonic component is identically zero for any set of scalar predictions.** This is because scalar values on nodes always define a consistent ranking with no cycles. The Hodge decomposition can only detect cycles in *exogenous* edge flows (human preferences), not in *endogenous* predictions from a scalar model.

This observation motivated our shift to the precomputed potential-alignment approach (Section ??), which uses the *global* Hodge decomposition of human preference data as an external regularization target.

## D. Full Pairwise Comparison Matrix

Table 4. Cohen’s  $d$  for all method pairs (30 seeds). Bold indicates  $p < 0.05$ .

|        | DPO | H-DPO       | GRPO        | H-GRPO      | KTO         | H-KTO       | ORPO        |
|--------|-----|-------------|-------------|-------------|-------------|-------------|-------------|
| DPO    | —   | <b>6.52</b> | <b>6.53</b> | <b>86.6</b> | <b>10.3</b> | <b>6.37</b> | <b>5.54</b> |
| H-DPO  |     | —           | <b>1.03</b> | <b>1.03</b> | <b>16.5</b> | <b>1.67</b> | <b>6.67</b> |
| GRPO   |     |             | —           | 1.03        | <b>16.5</b> | <b>1.67</b> | <b>6.67</b> |
| H-GRPO |     |             |             | —           | <b>86.6</b> | <b>1.03</b> | <b>6.67</b> |
| KTO    |     |             |             |             | —           | <b>16.5</b> | <b>2.77</b> |
| H-KTO  |     |             |             |             |             | —           | <b>6.57</b> |
| ORPO   |     |             |             |             |             |             | —           |